

TabPFN Predictive Oracles + Martingale Posteriors

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Project Vision

Bridging Foundational Models and de Finetti Theory

- **TabPFN** - foundational model for tabular data [1]
outperforming e.g. XGBoost (in fraction of time):
 - Inputs: $(X_{\text{train}}, Y_{\text{train}}, X_{\text{test}})$
 - Outputs: posterior predictive distribution
 $\pi(Y_{\text{test}} | X_{\text{test}}, X_{\text{train}}, Y_{\text{train}})$
 - Learns to perform model selection and parameter estimation automatically by amortizing over millions of synthetic datasets
 - Trained as a transformer meta-learner on diverse tabular tasks from a broad prior over generative processes using the in-context learning technique

Martingale Posteriors: Predictive Bayesian Updating

Classical Bayesian learning:

$$\pi(\theta|D) \propto p(D|\theta)\pi(\theta)$$

$$\pi(y_{\text{test}}|D) = \int \pi(y_{\text{test}}|\theta)\pi(\theta|D)d\theta$$

Suffers from misspecification and strong prior dependencies

Martingale posteriors [2] go the opposite way:

$$\pi(y_{\text{test}}|D) \xrightarrow[\text{Exchangeability}]{\text{Doob's theorem}} \pi(\theta|D)$$

by defining a sequence of one-step-ahead predictions

Idea: Predictive resampling: repeatedly simulate future observations $Y_{n+1}, \dots, Y_N$ from the predictive sequence, attach to observed data, compute θ on the completed dataset, and repeat

Martingale Posteriors: Predictive Bayesian Updating

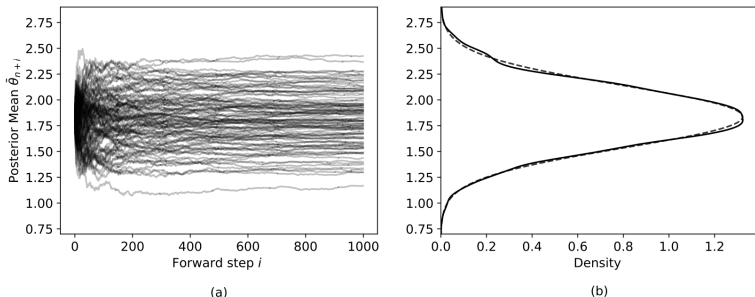


Figure 1: (a) Sample paths of $\bar{\theta}_{n+i}$ through forward sampling; (b) Kernel density estimate of $\bar{\theta}_N$ samples (—) and analytical posterior density $\pi(\theta | y_{1:n})$ (---)

- The empirical distribution of θ across resamplings approximates the posterior $p(\theta|D)$
- Ties predictive performance directly to posterior inference, bypassing explicit likelihoods and priors
- Offers robustness under misspecification

Bridging Predictive Oracles and Martingale Posteriors

Key idea: Use TabPFN's predictive distribution as the input to the martingale posterior framework

$$\pi_{\text{TabPFN}}(y_{t+1}|x_{t+1}, D_t), \quad M_t = \prod_{s=1}^t \pi_{\text{TabPFN}}(y_s|x_s, D_{s-1})$$

Result: A general framework for parameter inference without explicit likelihoods and priors, combining predictive power with theoretical coherence

Reworked de Finetti representation:

- Exchangeability $\rightarrow$ predictive sequence $\rightarrow$ posterior
- Links modern predictive models to Bayesian foundations

Paper Review: Nagler & Rügamer (2026)

Uncertainty Quantification for Prior-Data Fitted Networks using Martingale Posteriors

Thomas Nagler & David Rügamer
LMU Munich & MCML

Core Problem:

- PFNs approximate posterior predictive distribution $p(y|x, D_n)$
- But this **only provides point estimates** (conditional mean, quantiles)
- **No uncertainty quantification** for these estimates or for the parameters
- Cannot disentangle aleatoric vs. epistemic uncertainty

Proposed Solution: Construct Bayesian posteriors for PFN estimates using martingale posteriors through Copulas

The Problem: TabPFN Lacks the Martingale Property

Martingale property:

$$\mathbb{E}[p(y|y_{1:(n+k)})|y_{1:n}] = p(y|y_{1:n})$$

Problem:

- TabPFN **fails** the martingale check
- Without martingale property, MP sampling leads to meaningless (?) results

Authors' solution: Use PFN's PPD only as starting point, then update using nonparametric Gaussian copula ensuring martingale property

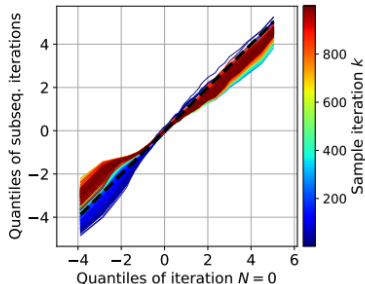


Figure 1: PFN's initial CDF vs. average CDF after forward iterations

Why Does the Martingale Property Matter?

From Emil's presentation:

If MP property holds: $F_{\infty, x}$ has well-defined distribution

Posteriors are valid Bayesian objects

If violated: $F_{\infty, x}$ may drift unpredictably

Uncertainty quantification is meaningless

TabPFN's issue:

- Trained to approximate PPD, not to maintain martingale consistency
- Forward sampling causes distribution to shift (recall QQ plot in Figure 1)

Classical MP: Unconditional Case (Fong et al. 2023)

Setup: No features, interest in unconditional parameter θ

Sampling scheme:

$$y_{n+1} \sim p(\cdot | y_{1:n}), \quad y_{n+2} \sim p(\cdot | y_{1:(n+1)}), \quad \dots$$

Predictive joint:

$$p(y_{(n+1):(n+N)} | y_{1:n}) = \prod_{i=1}^N p(y_{n+i} | y_{1:(n+i-1)})$$

Martingale posterior:

- Long-run empirical distribution:

$$F_{\infty}(y) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \mathbb{I}(y_{n+i} \leq y)$$

- F_{∞} is random, its distribution is $\Pi(F_{\infty} | D_n)$
- For parameter θ : $\Pi(\theta \in A | D_n) = \int \mathbb{I}(\theta(F_{\infty}) \in A) d\Pi(F_{\infty} | D_n)$

Extension to Conditional Case: Nagler & Rügamer's Innovation

Goal: Inference for **conditional** parameter $\theta(x) = \theta(P(\cdot|x))$

Examples:

- Conditional mean: $\theta(x) = \mathbb{E}[Y|X = x] = \int y dP(y|x)$
- Conditional quantile: $\theta(x) = P^{-1}(\alpha|x)$
- Conditional variance: $\theta(x) = \text{Var}(Y|X = x)$

Key insight: Fix x , sample **only labels** y conditional on $x_{n+k} = x$ for all $k \geq 1$

Conditional MP: Mathematical Formulation

Setup: Fix feature value x^* , observed data $D_n = \{(x_i, y_i)\}_{i=1}^n$

Augmented data: Set $x_{n+k} = x^*$ for all $k \geq 1$

Our predictive sequence becomes:

$$p_k(y) := p(y_{n+k+1} | D_n \cup \{(x^*, y_{n+1}), \dots, (x^*, y_{n+k})\})$$

Iterative sampling:

$$y_{n+1} \sim P_0, \quad y_{n+2} \sim P_1, \quad y_{n+3} \sim P_2, \quad \dots$$

where P_k is CDF of p_k

Conditional martingale posterior:

$$F_{\infty, x^*}(y) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \mathbb{I}(y_{n+i} \leq y)$$

Integrating over $\Pi(F_{\infty, x^*} | D_n)$ gives posterior for $\theta(x^*)$

The Challenge: Computing P_k Updates

Problem: After sampling $y_{n+1}, \dots, y_{n+k}$, need to compute:

$$P_k(y) = p(y | D_n \cup \{(x^*, y_{n+1}), \dots, (x^*, y_{n+k})\})$$

As we wanted too. But this is problematic:

- PFN gives $P_0 = CDF(y | D_n, x^*)$ easily (single forward pass)
- But PFN on $(D_n \cup \{(x^*, y_{n+1}), \dots, (x^*, y_{n+k})\})$ requires:
 - New forward pass through transformer: $O((n+k)^2)$ operations
 - For $k = 1, \dots, N$: total complexity $O(N^3)$ - **prohibitive!**
- Moreover, iterative PFN calls violate martingale property

Authors' solution:

- Use PFN only for P_0
- Update $P_0 \rightarrow P_1 \rightarrow P_2 \rightarrow \dots$ using **Gaussian copula** method
- This ensures martingale property + efficiency: $O(1)$ per update! So, **$O(N)$** per path simulation!

Gaussian Copula Updates: The Formula

Update rule: From P_{k-1} and new sample y_{n+k} , compute P_k as:

$$P_k(y) = (1 - \alpha_{n+k})P_{k-1}(y) + \alpha_{n+k}H_\rho(P_{k-1}(y), P_{k-1}(y_{n+k}))$$

where:

- $\alpha_{n+k} = C(n+k+1)^{-\beta}$ is **learning rate** (C, β hyperparameters)
- $H_\rho(u, v)$ is **Gaussian copula CDF**:

$$H_\rho(u, v) = \Phi \left(\frac{\Phi^{-1}(u) - \rho\Phi^{-1}(v)}{\sqrt{1 - \rho^2}} \right)$$

- Φ is standard normal CDF
- $\rho \in [-1, 1]$ is hyperparameter controlling correlation/smoothness

Weighted average of: Previous distribution P_{k-1} and copula-based update incorporating new data y_{n+k} .

Why Gaussian Copula? Three Key Reasons

Martingale property guaranteed

- Fong et al. (2023) proved: this update scheme satisfies

$$\mathbb{E}[P_k(y)|y_{1:n}] = P_{k-1}(y)$$

- Ensures coherent belief updating, no drift
- Foundation for all theoretical guarantees

Computational efficiency

- Each update: evaluate Φ , Φ^{-1} , arithmetic operations
- Complexity: $O(1)$ per update, total $O(N)$ for N samples
- Compare: iterative PFN calls would be $O(N^3)$
- Enables practical implementation with $N \sim 200$, $B \sim 100$ replications

Learning Rate Schedules: Why They Matter

Recall update formula:

$$P_k(y) = (1 - \alpha_{n+k})P_{k-1}(y) + \alpha_{n+k}H_\rho(P_{k-1}(y), P_{k-1}(y_{n+k}))$$

Role of α_{n+k} :

- Controls how much each new observation influences the distribution
- α_{n+k} large: rapid learning, high sensitivity to new data
- α_{n+k} small: slow learning, more stable estimates

Key requirement: Must decay to zero for convergence

$$\alpha_{n+k} \rightarrow 0 \text{ as } k \rightarrow \infty$$

But how fast? Depends on convergence rate of underlying estimator!

Three Learning Rate Schedules

General form: $\alpha_{n+k} = C(n+k+1)^{-\beta}$; d - number of covariates

Schedule	C	β
Default	2	1
Type 1	$10^{-(d-1)/(d+4)}$	$1/2 + d/(d+4)$
Type 2	$10^{-2d/(d+4)}$	$d/(d+4)$

Default (Fong et al. 2023):

- Assumes no bias, correct specification

Type 1:

- Assumes estimator is consistent (bias $\rightarrow 0$)

Type 2:

- Appropriate when **bias persists** (TabPFN case!)

Theoretical Justification: Proposition 3.3

Proposition 3.3 (Contraction rate): For any $y \in \mathbb{R}$, $N \geq 0$, with probability at least $1 - \delta$:

$$|P_{\infty, x}(y) - P_N(y)| \lesssim \sqrt{\frac{\log(2/\delta)}{2\beta - 1}} (n + N)^{-\beta + 1/2}$$

Three regimes:

1. $\beta > 1/2$: bound $\rightarrow 0$ as $N \rightarrow \infty$ (contraction)
 - Posterior concentrates around true value
 - Valid if estimator is consistent
2. $\beta = 1/2$: bound $\sim (n + N)^0 = O(1)$ (no contraction)
 - Posterior maintains constant uncertainty
3. $\beta < 1/2$: bound **increases** as $N \rightarrow \infty$
 - Posterior spreads out (very conservative)
 - Appropriate for severe misspecification

Theoretical Guarantees: Three Key Results

Proposition 3.1 (Exchangeability):

$$(y_{N+1}, y_{N+2}, \dots) \xrightarrow{d} (z_1, z_2, \dots)$$

where $(z_1, z_2, \dots)$ has exchangeable distribution

Proposition 3.2 (Convergence):

$$\lim_{N \rightarrow \infty} P_N(y) = P_{\infty, x}(y) = F_{\infty, x}(y) \text{ almost surely}$$

Proposition 3.3 (Contraction rate): With probability $\geq 1 - \delta$:

$$|P_{\infty, x}(y) - P_N(y)| \lesssim \sqrt{\frac{\log(2/\delta)}{2\beta - 1}} (n + N)^{-\beta+1/2}$$

Corollary 3.1: Importance of Initialization

Corollary 3.1: Let $P_{0,1}$ and $P_{0,2}$ be two initializations, with limits $P_{\infty,x,1}$ and $P_{\infty,x,2}$. Define $\Delta(y) := P_{\infty,x,1}(y) - P_{\infty,x,2}(y)$. Then with probability $\geq 1 - 2\delta$:

$$|\Delta(y)| \geq |P_{0,1}(y) - P_{0,2}(y)| - O(n^{-\beta+1/2})$$

Interpretation:

- Difference in martingale posteriors is **at least as large as** difference in initializations (up to contraction rate)
- Initialization is **dominant factor** in determining posterior
- Copula updates cannot “fix” a poor initialization

Practical implication:

- Quality of TabPFN's P_0 is **crucial**
- If TabPFN gives poor predictive distribution, MP will be poor!

Algorithm 1: Complete Procedure

Algorithm 1 Martingale Posterior Computation

- 1: **Input:** Estimated $\widehat{\text{PPD}}(y|x)$ from PFN
 - 2: **for** $b = 1, \dots, B$ **do**
 - 3: Initialize $P_0^{(b)} \leftarrow \widehat{\text{PPD}}(y|x)$
 - 4: **for** $k = 1, \dots, N$ **do**
 - 5: Sample $y_{n+k}^{(b)} \sim P_{k-1}^{(b)}$
 - 6: Update $(P_{k-1}^{(b)}, y_{n+k}^{(b)}) \rightarrow P_k^{(b)}$ via Gaussian copula
 - 7: **end for**
 - 8: Compute $\hat{P}_N^{(b)}(y) = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(y_{n+i}^{(b)} \leq y)$
 - 9: Set $\theta^{(b)}(x) \leftarrow \theta(\hat{P}_N^{(b)})$ (e.g., mean, quantile)
 - 10: **end for**
 - 11: Compute estimated MP: $\hat{\Pi}(\theta(x)) \in A|D_n) = \frac{1}{B} \sum_{b=1}^B \mathbb{I}(\theta^{(b)}(x) \in A)$
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Convergence

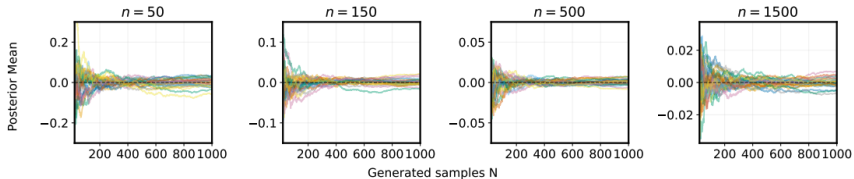


Figure 2: Convergence of conditional means on concrete data for different initial n and MP steps N

Simulation Study: Setup and Goals

Bayesian Additive Model:

$$Y|x, \theta \sim N \left(\sum_{j=1}^J f_j(x_j), \sigma^2 \right)$$

$$f_j(x_j) = B(x_j)^\top \theta_j, \quad \theta_j \sim N(0, I_{20})$$

where $B(x_j) \in \mathbb{R}^{20}$ is B-spline basis (degree 3, 20 functions)

Design:

- 30 features total
- $\phi \in \{1, 3, 10\}$ features with non-zero signal
- Remaining features: pure noise
- Sample sizes: $n \in \{30, 100, 300\}$
- Fixed variance: $\sigma^2 = 0.5$

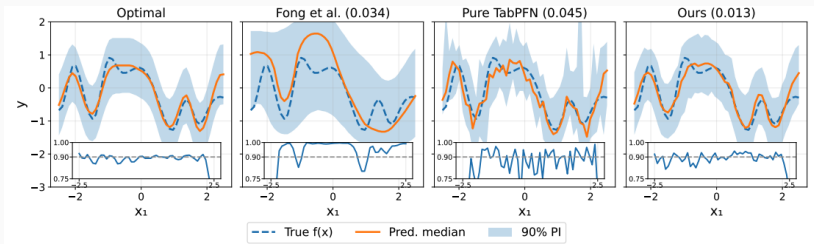


Figure 3: Comparison of estimated median function (orange line) and credible intervals (shaded blue area) for a non-linear feature effect (dashed blue line) in a Bayesian additive model based on different methods (facets). Numbers in brackets after each method state the marginal miscoverage, i.e., the absolute difference between the method's average coverage and the nominal level of 90% (smaller is better).

Simulation Results: Coverage Analysis

n	φ	Optimal	Fong et al.	Ours	Pure TabPFN
30	1	0.91 (1.99)	0.93 (2.57)	0.90 (2.12)	0.83 (2.60)
30	3	0.74 (2.84)	0.65 (3.09)	0.88 (3.52)	0.87 (3.56)
30	10	0.82 (5.57)	0.99 (6.64)	0.94 (5.83)	0.74 (5.91)
100	1	0.92 (1.85)	0.85 (2.65)	0.89 (1.83)	0.82 (1.92)
100	3	0.93 (2.02)	0.96 (3.54)	0.92 (2.77)	0.87 (2.94)
100	10	0.99 (3.17)	0.05 (1.05)	0.99 (6.12)	0.94 (5.83)
300	1	0.91 (1.76)	0.58 (1.54)	0.90 (1.78)	0.84 (1.75)
300	3	0.91 (1.83)	0.92 (3.09)	0.90 (2.16)	0.67 (2.13)
300	10	0.95 (2.06)	0.93 (5.04)	0.97 (5.10)	0.94 (5.06)

Figure 4: Average conditional coverage across different (n, ϕ) combinations

Real-World Benchmarks: UCI Datasets

Task: Conditional quantile regression, $\alpha \in \{0.5, 0.9\}$

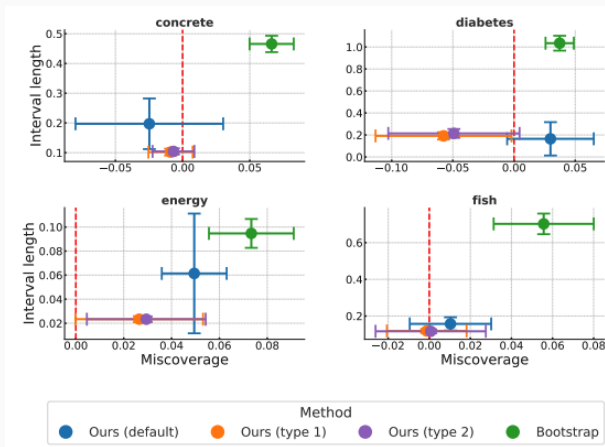
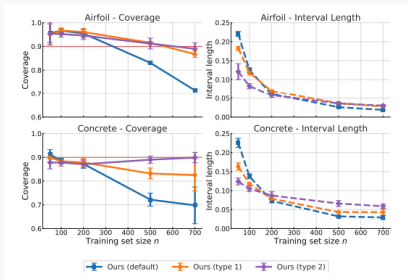


Figure 5: Miscoverage vs. interval length. Red line = perfect coverage (0.9). $n = 100$, $B = 100$, $N = 1000$?

Ablation Study: Dataset Size Effects

Investigation (UCI subset):

- Vary $n \in \{50, 100, 200, 500, 700\}$



- **Default & Type 1:** Coverage drifts away from 0.9 as n increases
- **Type 2:** Maintains coverage near 0.9, intervals shrink appropriately

Ablation Study: Initialization Importance

Question: How crucial is TabPFN initialization?

Experiment: Replace TabPFN P_0 with Fong et al.'s PPD

- Keep copula updates, bandwidth, learning rate identical
- Only change: initial distribution

	Default	Type 1	Type 2	Pure TabPFN
airfoil	0.18 (0.29)	0.05 (0.07)	0.12 (0.19)	0.27 (0.68)
boston	0.25 (0.23)	0.08 (0.08)	0.19 (0.18)	0.27 (0.51)
concrete	0.16 (0.30)	0.05 (0.08)	0.11 (0.21)	0.34 (1.02)
diabetes	0.34 (0.29)	0.12 (0.10)	0.27 (0.23)	0.65 (0.79)
energy	0.27 (0.12)	0.09 (0.04)	0.20 (0.09)	0.53 (0.89)
fish	0.24 (0.22)	0.06 (0.06)	0.17 (0.15)	0.54 (0.70)

Figure 6: Coverage with Fong initialization, $N = 1000$

Result: Dramatic coverage degradation

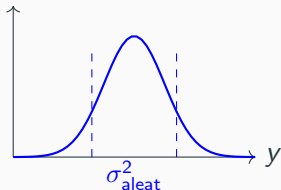
Confirms Corollary 3.1: Initialization dominates posterior quality!

Separating Aleatoric and Epistemic Uncertainty Separation

Aleatoric Uncertainty

- Irreducible randomness in $Y|X = x$
- Captured by *spread* of $P_{n+k}(y|x)$
- Does **not** decrease with n
- Intrinsic noise in data

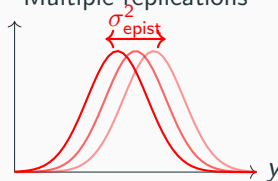
Single replication



Epistemic Uncertainty

- Parameter uncertainty
- Captured by *variance across replications*
 $b = 1, \dots, B$
- **Contracts** as $n \rightarrow \infty$
- Reducible with more data

Multiple replications





Noah Hollmann et al.

Accurate predictions on small data with a tabular foundation model.

Nature, 637(8045):319–326, 2025.



Edwin Fong, Chris Holmes, and Stephen G Walker.

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Pre-tuned untrained predictors.

Technical report, 2023.



Steven R Howard et al.

Time-uniform Chernoff bounds via nonnegative supermartingales.

Probability Surveys, 17:257–317, 2020.

Thank you!

Questions & Discussion